

Paul's Online Notes

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Section 13.5 : Differentials

This is a very short section and is here simply to acknowledge that just like we had **differentials** for functions of one variable we also have them for functions of more than one variable. Also, as we've already seen in previous sections, when we move up to more than one variable things work pretty much the same, but there are some small differences.

Given the function $z = f(x, y)$ the differential dz or df is given by,

$$dz = f_x dx + f_y dy \quad \text{or} \quad df = f_x dx + f_y dy$$

There is a natural extension to functions of three or more variables. For instance, given the function $w = g(x, y, z)$ the differential is given by,

$$dw = g_x dx + g_y dy + g_z dz$$

Let's do a couple of quick examples.

Example 1 Compute the differentials for each of the following functions.

(a) $z = e^{x^2+y^2} \tan(2x)$

(b) $u = \frac{t^3 r^6}{s^2}$

(a) $z = e^{x^2+y^2} \tan(2x)$ **Hide Solution ▼**

There really isn't a whole lot to these outside of some quick differentiation. Here is the differential for the function.

$$dz = \left(2xe^{x^2+y^2} \tan(2x) + 2e^{x^2+y^2} \sec^2(2x) \right) dx + 2ye^{x^2+y^2} \tan(2x) dy$$

(b) $u = \frac{t^3 r^6}{s^2}$ [Hide Solution](#) ▼

Here is the differential for this function.

$$du = \frac{3t^2 r^6}{s^2} dt + \frac{6t^3 r^5}{s^2} dr - \frac{2t^3 r^6}{s^3} ds$$

Note that sometimes these differentials are called the **total differentials**.